

Lemma F — the antipodal freeness

PLANETVORTEX — the point monograph

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Statement

The witness involution a (order 2) acts on the 24 faces gZ without fixed points; hence it pairs them into 12 antipodal pairs — one pair per gravimetric register of stage V2.

Proof

Suppose a fixes the face gZ : $agZ = gZ$, hence $g^{-1}ag \in Z = \langle ab \rangle$. The left side has order 2 (conjugation preserves order), the right side is a group of order 7 — an element of order 2 cannot lie in a group of order 7. Contradiction. A fixed-point-free involution on a 24-element set is a product of 12 transpositions. $\square$

Computational certificate

Stage V2: the 12 antipodal pairs; `face_fixed_points` = 0 in the C99 kernel report as well.